

GlideinWMS Status Update

- Production version: 3.10.15
- Development version: 3.11.1 (to be released early next week)

Highlights from the Last Semester

- Better automation of Factory configuration:
 - • Management of HTCondor tarballs
 - • Improved site configuration management
- Custom per-Glidein log reporting and dedicated log server:
 - • Works even when Glideins are killed
 - • Separate logs for HPC multi-node jobs

Apptainer Integration

- Use of Apptainer distributed via HTCondor
- Improved environment management (preserved vs discarded variables)
- Added GLIDEIN_Name and GLIDEIN_UUID for CMS jobs (HammerCloud)

Platform and Compatibility Enhancements

- Easier Glidein HTCondor configuration
- Support for Cvmfsexec mode 1 (mountrepo)
- Support for Ubuntu 24 worker nodes
- Support for jobs using Alpine Linux-based containers

Credential and Code Improvements

- New credential library (development version):
 - Generators for SciTokens and IdTokens
 - Credential caching and dynamic per-site credentials
- Modernized codebase (typing, removed M2Crypto dependency)

Looking Ahead

- With the WMAgent freeze and the design of a new WM system, future needs are evolving.
- Several plans and goals are emerging for GlideinWMS and Submission Infrastructure.

Main Goals (Planned Work)

- CGROUPS v2 support to limit memory, CPU, and resource usage (EL9 nodes)
- Condor CONCURRENCY LIMITS to restrict I/O-heavy jobs and protect site resources

CVMFS and Containerization

- Full CVMFSEXEC support (Portuguese HPC and others):
 - Mode 1 (mountrepo) already supported
 - Mode 3 (user namespaces) coming soon
- Native HTCondor invocation of Singularity (enables condor_ssh_to_job)

Scalability and Credential Enhancements

- New credential generators (in development, need hardening)
- Uber-glidein and scalability tools improvements
- Allow sites to advertise custom attributes (ClassAds) for improved matchmaking

Glidein Feedback Loop

- Glideins provide runtime info back to the Factory to enrich configuration:
 - • GPU availability and specs
 - • Microarchitecture and arch details
 - • Node micro-benchmark results

Future Directions: HPC Integration

- HPC support remains a key focus
- Improve automation of SSH-based submissions (HEPCloud)
- Integrate CMS 'glideins in a vacuum' into GlideinWMS
- Containerized wrappers for easier deployment (based on student work)

Architectural Improvements

- Decouple Factory and schedd components
- Simplifies containerized deployments and improves flexibility

Next Steps

- Validate priorities with PPD and CompOps
- Get feedback from Kevin before expanding scope
- Continue development toward next GlideinWMS release